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A review of vertical drill-string mathematical modelling

Jasem M. Kamel [a,*], Asan G.A. Muthalif [b], Abdulazim H. Falah [c]

[a] Kuwait National Petroleum Company, Kuwait
[b] Mechanical and Industrial Engineering, Qatar University, Qatar
[c] Mechanical Engineering Department, Kuwait University, Kuwait

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ABSTRACT

This work aims to elucidate the fundamental concepts and approaches for modelling the vertical rotary drilling system and analysing drill string vibrations through a comprehensive literature review. It offers a detailed examination and critical discussion for the discrete and distributed mathematical modelling vibration along with associated boundaries conditions for low and high frequency oscillations. **Additionally, this study discusses experimental models and their actual applications, highlighting their role in validating theoretical models and improving drilling performance.** This study presents a thorough overview of existing literature on vertical drill string vibration problems, making it a valuable resource for researchers in the field. The study not only synthesizes existing knowledge but also seeks to guide future research efforts in addressing and mitigating the complex challenges associated with drill string vibrations.

1. Introduction

The global demand for oil and natural gas extraction through deep borehole drilling is increasing due to its essential role in various energy dependent industrial sectors. However, this process encounters significant challenges, including severe vibrations. These vibrations are self-excited and can be categorized to three main modes: torsional, lateral and axial. They are primarily observed at the bottom hole and often occur with huge amplitudes. Specifically, axial vibrations mode manifest as bit bouncing and High Frequency Axial Oscillations (*HFAOs*), torsional vibrations mode as stick slip and High Frequency Torsional Oscillations (*HFTOs*), and lateral vibrations mode as whirling. These forms of vibrations modes are among the most destructive encountered during drilling operations (Dareing, 1984; Ghasemloonia et al., 2015; Zhu et al., 2014; Shen et al., 2021; Zhang et al., 2017). The fundamental vibrations modes along with high frequency vibrations modes are self-excitation, coupled (Li et al., 2021; Zhang et al., 2023; Christoforou and Yigit, 2003) and serve as excitation sources to each other (Besaisow and Payne, 1988; Richard et al., 2004; Sugiura and Jones, 2019). which results in the extremely complex dynamic behaviors of drill string (Chen et al., 2020). The main sources of these vibration modes are numerous, including drill string resonance at its natural frequency, the bit and the formation interactions, characteristics of the drilled Rocks, bottom hole environmental conditions, impacts between multiple points along the drill string and the borehole wall, mass imbalances, properties of the drilling fluid, improper operational parameters, and factors such as initial curvature in drill collars and vibration-inducing tools. Additionally, the structural nature of the drilling system plays a significant role, particularly when the system has a large slenderness ratio and minimal stiffness, making it prone to plasticity oscillations. The consequence of occurrence of these vibrations during drilling are detrimental to the drilling system components (Albdiry and Almensory, 2016), that lead to drill string and bottom hole assembly (*BHA*) damage, bit wear, reduced penetration rates, abrasive wear of tubulars, drilling inefficiency, high maintenance, and operational cost (Global Oil et al., 2024) and loss of time. Bottom Hole Assembly tools generally fail in two ways (Macdonald and Bjune 2007). One way occurs when the vibration frequency approaches the natural frequency of the drill string, resulting in resonance. This resonance causes the response value to reach or exceed the ultimate strength of the drill string components, leading to rupture. The other way involves fatigue damage in the drill string components due to prolonged or repetitive vibrations. Fatigue is the primary cause of drill string failure where 72 % of the *BHA* fractur failure belongs to fatigue fracture (Dong and Chen, 2016; Baryshnikov et al., 1997). Extensive research has been undertaken to comprehensively understand these phenomena, driven by field observations and the detrimental impacts they cause. The goal is to optimize drill string design and drilling operation parameters, facilitating smoother drilling processes and

* Corresponding author.
E-mail addresses: ja1808926@qu.edu.qa (J.M. Kamel), drasan@qu.edu.qa (A.G.A. Muthalif), Abdulazim.falah@ku.edu.kw (A.H. Falah).

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minimizing or eliminate the unwanted vibration effects. Many measures have been taken and One of the effective proactive approaches to address these challenges and minimize costs is through mathematical modeling of drilling systems. This complex task plays a crucial role in improving drilling efficiency and safety by accurately predicting drill string behaviour and assessing the impact of operational and structural parameters. For instance, recent studies have utilized cascaded dynamic models to enhance fatigue life prognosis in drill strings, enabling better maintenance strategies and performance optimization (Galagedarage Don and Rideout, 2023). By identifying the root causes of encountered problems, engineers can develop appropriate solutions to avoid detrimental vibration effects and enhance the drilling system's performance.

This paper aims to provide a comprehensive literature review on the proposed rotary drill string mathematical modeling approaches Lumped parameter and Distributed models along with associated boundary conditions for low and high frequency oscillations and offers a deep discussion on the previous studies showing gaps and future recommendations for research. **Additionally, the study discusses experimental models and their actual applications, emphasizing their role in validating theoretical models and enhancing drilling performance.**

2. Mathematical modeling of drill string

The mathematical modeling of drilling systems is a crucial and a complex task that is considered as an effective tool for improving the efficiency and safety of rotary drilling operations by predicting the drill string behavior and identifies the effects of regulating the surface drilling operation parameters on the drill string behavior, Operation Parameters such as torsional and axial speeds, torque, and Weight on Bit in addition to vibration generator tools, (Chin, 1994; Kamel and Yigit, 2014). That assists the site engineers to identify the potential problems and develop a proper solution to improve the performance of the drilling system and avoid detrimental vibrations problems. Numerous dynamic models were proposed considered the consequences of coupled vibrations in drill strings in torsional, axial, and lateral directions (Christoforou and Yigit, 2003; Kamel and Yigit, 2014; Richard et al., 2007; Yigit and Christoforou, 1998). The purposes of creating these dynamic models and motivations for enhancing them were to be able to predict the drill string behavior and to identify occurrence of the undesirable harmful vibrations (like bit bouncing, stick-slip, and whirling), optimizing the operation drilling parameters, estimate the exact critical rotary speeds, understand the drill string's transient, steady state response, and contact behavior, determine the reaction forces at the surface and estimate the cutting forces at the bit.Which will serve to establish a corrective measure for managing violent vibration degrees in the field and to lay the groundwork for creating control algorithms to mitigate these vibrations. Three categories' classifications could be made from the studies in the past: creating active or passive control methods to reduce unwanted vibrations, theoretical analysis, experimental and field data investigation. Laboratory measurements (Real et al., 2018; Roohi et al., 2022; Sassi et al., 2021; Tian et al., 2021) field measurements (Ledgerwood et al., 2013; Li et al., 2021) and simulations with well explained Finite Element model (Khulief and Al-Naser, 2005; Spanos et al., 1995; Tran et al., 2019) and lumped parameter model (Christoforou and Yigit, 2003; Kamel and Yigit, 2014; de Moraes and Savi, 2019) or continuous parameter models (Aarsnes and van de Wouw, 2019; Christoforou and Yigit, 1997; Tucker and Wang, 1999) are parts of the studies. Modelling has two different approaches. Some researchers use distributed parameter modelling (Akl et al., 2021) whereas a lumped parameter approach is adopted by other groups of researchers (Choe et al., 2023). Conversely, models to examine the drill string's vibration pattern may be developed in time (Hakimi and Moradi, 2009) or in frequency domain (Kamel and Yigit, 2014; Khulief and Al-Naser, 2005).

2.1. Lumped parameter model

The early study conducted in 1982 by Belokobyl'skii and Prokopov (Belokobyl'skii and Prokopov, 1982) marked a significant advancement in the understanding of rotary drilling dynamics. They identified that self-excited torsional vibration, primarily prompted by the frictional force between the drilling bit and formation during the rock breaking process. In their research, the drill string was modelled as a slender rod with rigid body at the top rotating at a constant torsional speed, while the bottom section was attached to a rigid cylinder and exposed to a torque created by friction. One of the key contributions of this paper was the establishment of a 1° of Freedom, torsional pendulum model with centralized mass. This model later became a widely accepted approach in the analysis of drilling dynamics. The authors capture the friction exists between the bit and the formation during breaking the rocks as Coulomb friction, a fundamental model in understanding frictional forces. They developed equations for the stick-slip vibration and calculated the period of these vibrations. This study is recognized from the early investigations into the stick-slip vibration phenomena in drill strings. However, it's important to note that this model did not take into account the effect of drilling mud viscous damping, which can be a significant factor in the dynamics of drilling operations. The exclusion of this factor means that while the study laid foundational groundwork, it only presents a partial view of the complex interactions within drill string systems. In same manner recently a simplified 1-degree torsional pendulum model was formulated by (Tang and Zhu, 2020), as illustrated in Fig. 1 of their publication. This model explores the impact of drill string length on the existence of stick-slip oscillations. The findings indicate a critical length factor that influences the occurrence of stick-slip. Furthermore, the study demonstrates that as the drill string length increases, there is a corresponding increase in the period of the stick phase, the torsional amplitude, and the period of the slip phase which reveals the fact that drill string geometric role on its response and the high complexity due to the counties changing in the drill system characteristics during operation.

Rudat and Dashevskiy (2011) developed a torsional pendulum model 1 DOF as illustrated in Fig. 2 and equation1 to capture the behaviour of the drill string, particularly when stick-slip phenomena occur, in order to reduce the harmful effects of stick-slip vibrations by optimizing drilling operation parameters. (Omojuwa et al., 2011) used Rudat's and Dashevskiy model to further study stick slip phenomena and found that

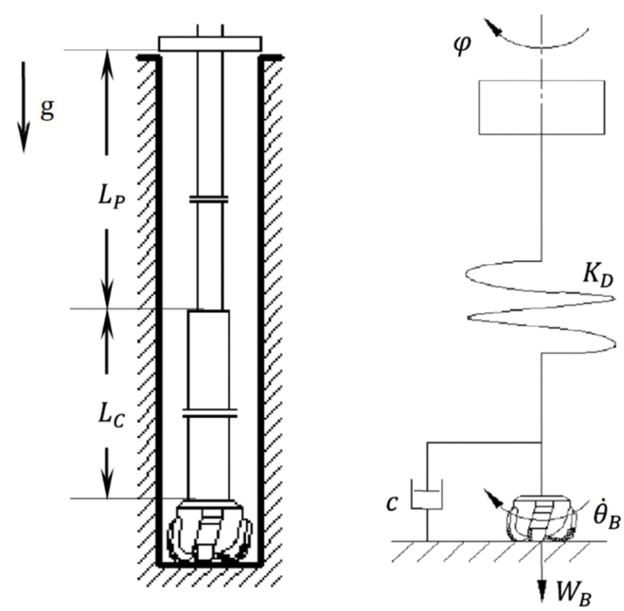

Fig. 1. the drill string mechanical model (Tang and Zhu, 2020).

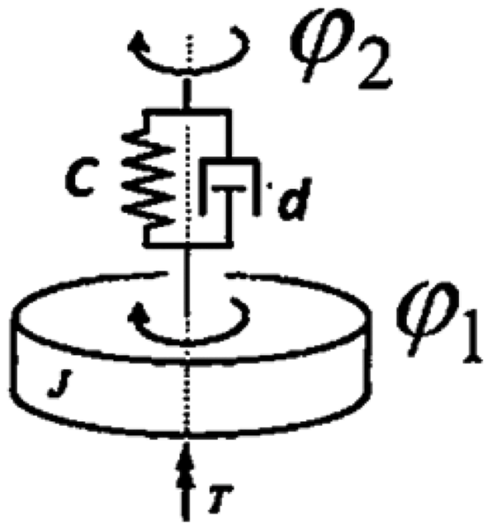

Fig. 2. drill string 1° of freedom mechanical model (Rudat and Dashevskiy, 2011).

increasing the *RPM* and reducing the Weight on Bit *(WOB)* will reduce the stick slip occurrence, but the latter one will reduce the Rate Of Penetration (*ROP)*. On other hand, the author approved that the stick slip vibrations of drill string could be reduced efficiently when using viscous damped environment as torsional vibration absorber as illustrated in Figs. 3 (a&b).

$$J\ddot{\varphi}_1 + d\dot{\varphi}_1 + c\varphi_1 = d\dot{\varphi}_2 + c\varphi_2 - T \tag{1}$$

where the angle, φ_1, is the torsional displacement of the rigid body, The φ_2 is a torsional position of the spring, *J, c, d,* and *T* are the mass moment of inertia of the rigid body, torsional spring and damper coefficients and Torque on Bit (*TOB)* respectively.

A lumped parameter model for describing coupled torsional and bending vibrations was developed by (Yigit and Christoforou, 1998), in which the nonlinear equations of motion equations 2 are obtained through a simplified model as shown in Fig. 4.

$$\begin{gathered}
(\boldsymbol{m}+\boldsymbol{m_f})(\ddot{\boldsymbol{r}}-\boldsymbol{r}\dot{\boldsymbol{\theta}}^2)+\boldsymbol{k}(\phi)\boldsymbol{r}+\boldsymbol{c_h}|\boldsymbol{v}|\dot{\boldsymbol{r}}=(\boldsymbol{m}+\boldsymbol{m_f})\boldsymbol{e}_0\left[\dot{\phi}^2\boldsymbol{cos}(\phi-\boldsymbol{\theta})+\ddot{\phi}\boldsymbol{sin}(\phi-\boldsymbol{\theta})\right]-\boldsymbol{F_r},\\
(\boldsymbol{m}+\boldsymbol{m_f})\left(\boldsymbol{r}\ddot{\boldsymbol{\theta}}+2\dot{\boldsymbol{r}}\dot{\boldsymbol{\theta}}\right)+\boldsymbol{c_h}|\boldsymbol{v}|\boldsymbol{r}\dot{\boldsymbol{\theta}}=(\boldsymbol{m}+\boldsymbol{m_f})\boldsymbol{e}_0\left[\dot{\phi}^2\boldsymbol{sin}(\phi-\boldsymbol{\theta})-\ddot{\phi}\boldsymbol{cos}(\phi-\boldsymbol{\theta})\right]+\boldsymbol{F_\theta},\\
\boldsymbol{J}\ddot{\phi}+\boldsymbol{k_T}(\phi-\phi_{\boldsymbol{rot}})+\boldsymbol{c_v}\dot{\phi}+\boldsymbol{c_h}|\boldsymbol{v}|\dot{\boldsymbol{r}}\boldsymbol{e}_0\boldsymbol{sin}(\phi-\boldsymbol{\theta})-\boldsymbol{c_h}|\boldsymbol{v}|\boldsymbol{r}\dot{\boldsymbol{\theta}}\boldsymbol{e}_0\boldsymbol{cos}(\phi-\boldsymbol{\theta})\\
=-\boldsymbol{T}(\phi)+\boldsymbol{F_\theta}[\boldsymbol{R}-\boldsymbol{e}_0\cos(\phi-\boldsymbol{\theta})]-\boldsymbol{F_r}\boldsymbol{e}_0\sin(\phi-\boldsymbol{\theta}).
\end{gathered} \tag{2}$$

where ϕ_{rot} is the angular displacement at the rotary table, ϕ is the angle

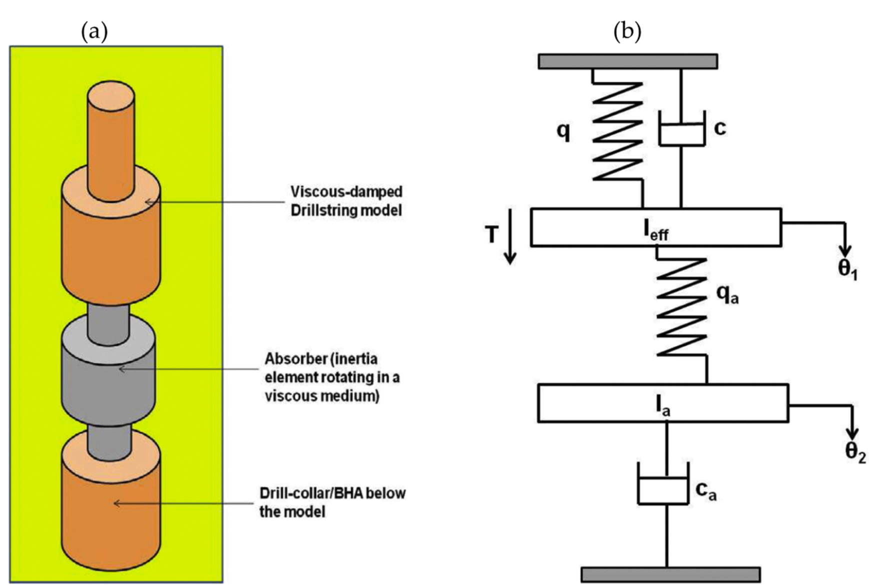

Fig. 3. (a); Torsional vibration absorber, (b) Torsional pendulum system with absorber (Omojuwa et al., 2011).

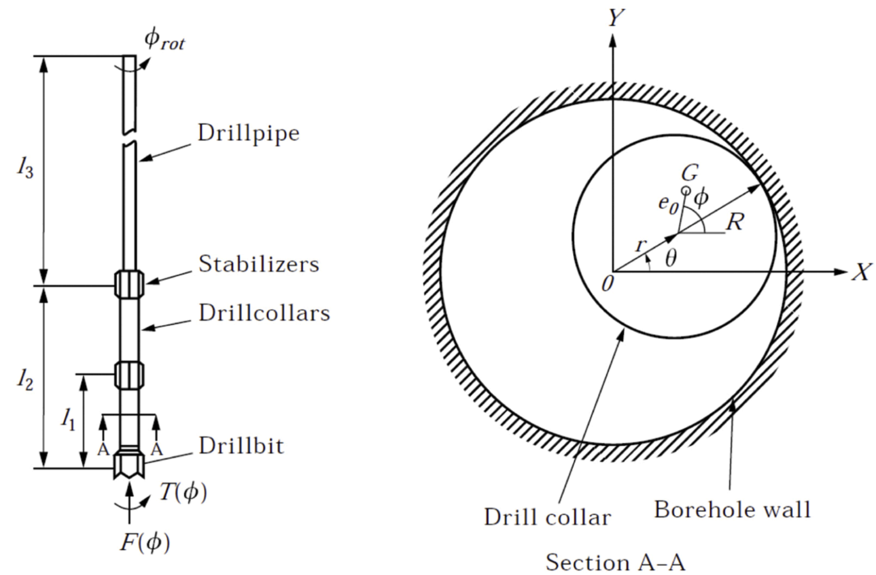

Fig. 4. The sketch of the system (Yigit and Christoforou, 1998).

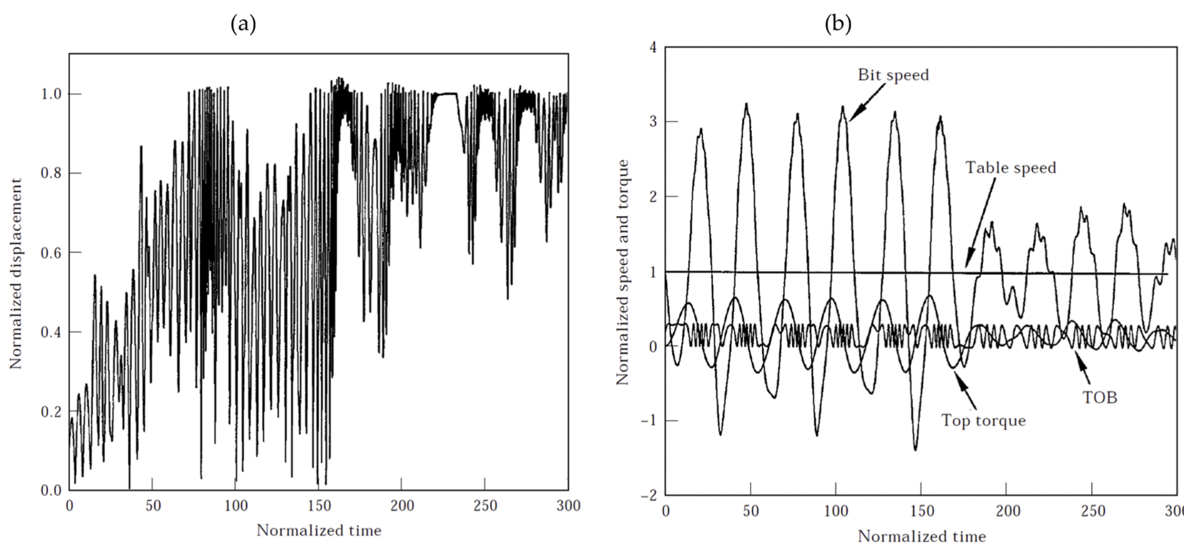

Fig. 5. (a) Lateral vibrations response during borehole wall impacts (b)Torsional vibrations response during borehole wall impacts (Yigit and Christoforou, 1998).

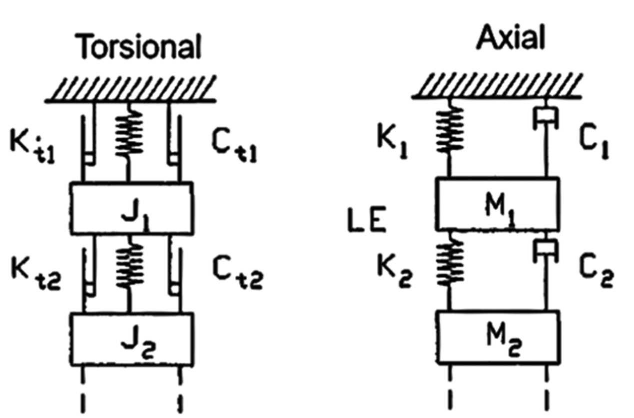

Fig. 6. the drill string simplified axial and torsional model (Elsayed et al., 1997).

of rotation of the drill collars, r is the radial displacement, v is the velocity of the geometric center of the drill collar section, e_o is the eccentricity of the center of mass with respect to the geometric center of the collar section. The parameters J, m, m_f, k, k_T, c_h, and c_v are defined as follows: J is the equivalent mass moment of inertia, m is the mass, m_f is the added fluid mass, k is the transverse stiffnessk_T, is the torsional stiffness, c_h, and c_v are the hydrodynamic and viscous damping coefficients, respectively. R is the radius of the drill collars. $T(\phi)$ is the torque on bit (TOB) and $F(\phi)$ is the weight on bit (TOB). F_θ *and* F_r are tangential and radial drill sting and bore hole wall interaction forces.

The equivalent lumped parameters generated from related distributed model of the drill string with a one mode approximation for coupled lateral and torsional vibrations. The rotation of the bit is related to the excitation resulting in completely coupled behavior. The rolling and slip-rolling behaviors of *BHA* could both be captured and analyzed using the suggested model. The lateral and torsional nonlinear coupling effects the responses and instabilities occur at critical rotary speeds, which may not be identified as crucial by previous uncoupled model. There is a notable energy transfer observed between the two vibrations

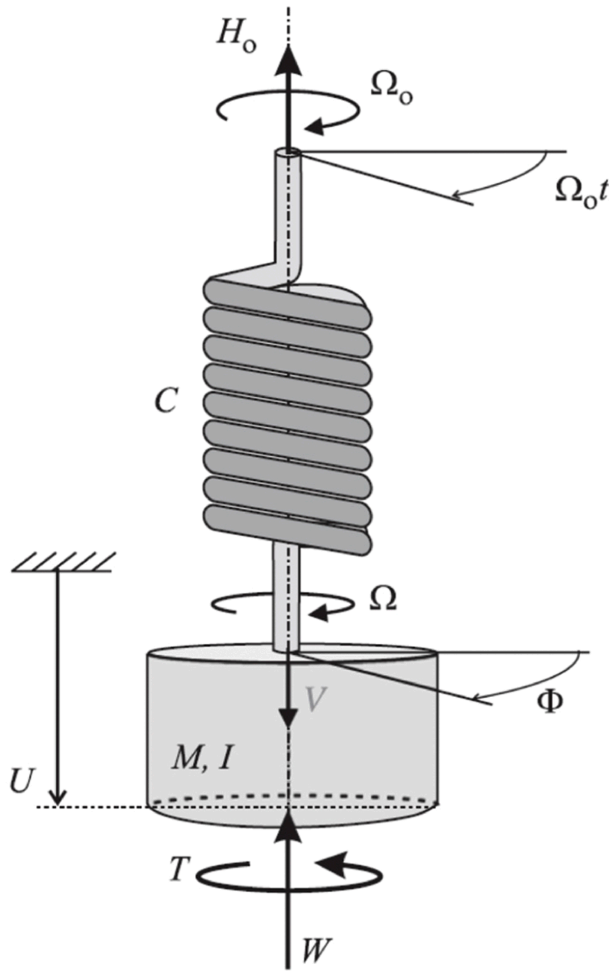

Fig. 7. the drilling system simplified model (Richard et al., 2007).

modes at specific frequencies as indicated in Figs. 5 (a and b).

Drill string torsional and axial vibrations model with a multi DOF system as illustrated in Fig. 6 was analyzed by (Elsayed et al., 1997) to investigate the effects of torsional vibrations on the drill string overall dynamic behavior and concluded that the torsional motion is highly to be considered in drilling system analysis moreover they attained stability conditions for stick-slip oscillations.

(Richard et al., 2004; Richard et al., 2007) researched upon the self-excited stick slip vibrations of a vertical drill string with a *PDC* bit, employing a lumped parameter model that considers the torsional and longitudinal modes of vibration of the drill bit as illustrated in Sketch Fig. 7. The vibration modes are coupled by the bit and rock interaction law suggested by (Detournay and Defourny, 1992) which takes into account frictional contact as well as cutting reaction at the bit/rock interface. The cutting forces are generated as function of the depth of cut which is measured by the variations in the bit axial response at successive bit rotations. On the other side, the frictional forces are assumed to be mobilized along the wear flats and rate independent. Hence, owing to the delayed, coupled and discontinuous bit-rock interaction, the equations of motion are nearly nonlinear. According to the results, the occurrence of self-excited vibrations, resulting in stick slip oscillations under specific conditions is greatly linked with the delay in the equations of motion.

In order to evaluate the torsional and axial self-excited drilling system vibrations equipped with *PDC* bit, (Germay et al., 2009) investigated the methodology suggested by (Richard et al., 2007). This was done by formulating a continuous model to represent the drill string behavior instead of characterizing the drilling system by a 2° of freedom model composed by a punctual mass, torsional inertia, and torsional stiffness value. The drilling structure's many natural torsional and axial vibration modes were captured using the continuum model. The boundary conditions at the *PDC* bit/rock interface considered the cutting and frictional reactions. A regenerative effect occurs through the cutting process which is associated with the quasi helicoidal bit motion causing axial and torsional vibrations modes coupling along with delay depending in the equations of motion, Whereas the frictional contact mechanism causes discontinuities in the bit boundary conditions when the bit becomes stuck in its angular and axial motion. finite element method is used to compute the drilling system dynamic response (see Eq. (3)).

$$\rho J\frac{\partial^2\Phi}{\partial t^2} - GJ\frac{\partial^2\Phi}{\partial x^2} = f_\Phi,\quad \rho A\frac{\partial^2 U}{\partial t^2} - EA\frac{\partial^2 U}{\partial x^2} = f_U \tag{3}$$

The angular (axial) displacement is represented by $\Phi(U)$, the material density is defined by ρ, the drill string material's Young's (shear) modulus is expressed by *E(G),* the cross-section area and polar moment of inertia are represented by A and J respectively, the axial body force due to gravity is represented by f_u, and the body force due to shear stresses between the pipe and the mud is represented by f_Φ. The study carried out by (Germay et al., 2009) was further developed by (Besselink

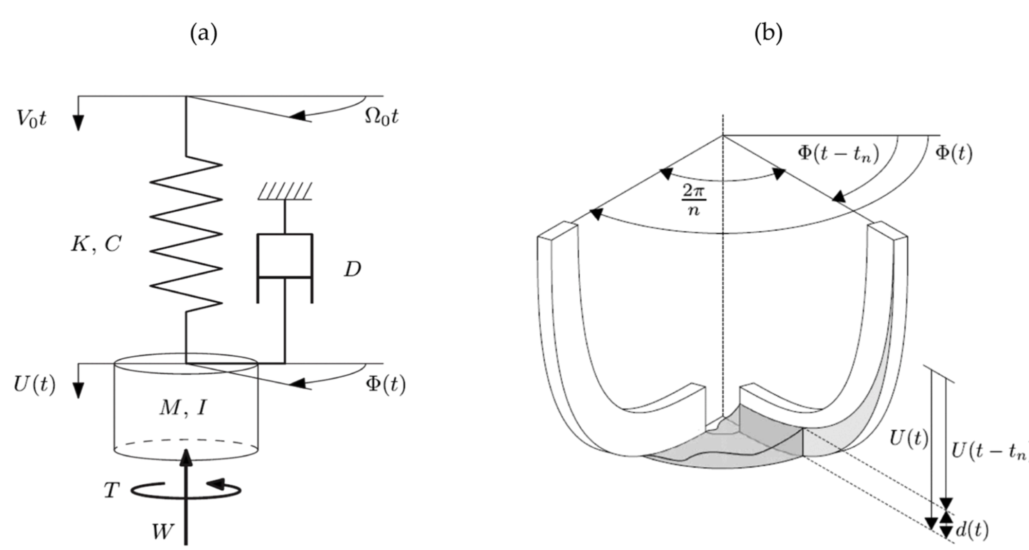

Fig. 8. (a) Drill string Schematic model; (b) Bottom hole profile between two consecutive blades (Germay et al., 2009).

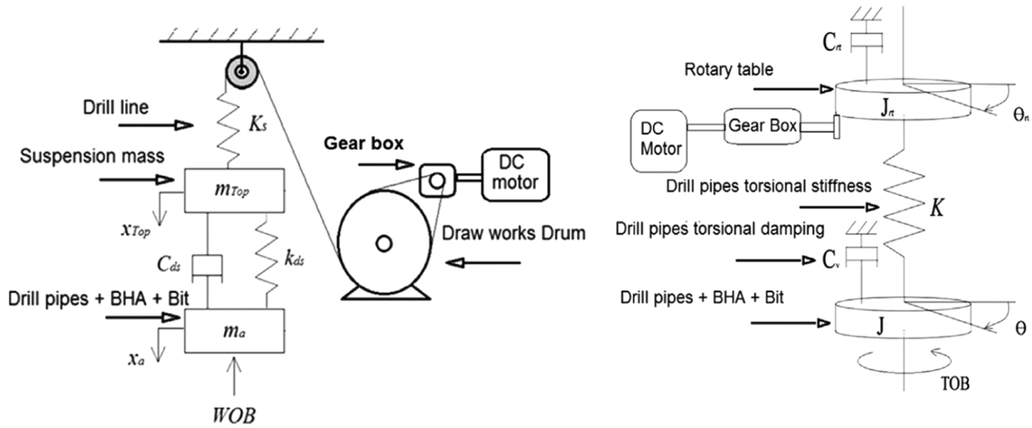

Fig. 9. Axial and Rotational models of the simplified proposed system (Kamel and Yigit, 2014).

et al., 2011). The coupling through employing a rate independent bit/rock interaction law between the axial and torsional vibrations was considered in their model. Two major characteristics were included to modify this model which are axial viscous friction taking the role of the dissipation along the bottom hole assembly denoted as (D) and the axial stiffness which shows the drill string axial flexibility denoted as (K) in Eq. (4) and Fig. 8(a) in addition the author introduced an instantaneous depth of cut measuring method that depends on the current axial position of the bit cutter with respect to the rock profile, as generated by the previous blade at time t_n ago. This is schematically depicted in Figure 8 (b). Therefore, the rock height in front of a single bit blade is described by the depth of cut (*DOC*). For estimating the exact axial limit cycle, a semi-analytical approach was used which employs different time scales related to the torsional and axial dynamics. Simulation results show that reduction in the torsional vibrations can be brought through the stabilization of the axial vibrations.

$$\begin{aligned} &M\frac{d^2U}{dt^2} + D\frac{dU}{dt} + K(U - V_0 t) = -W^c - W^f \\ &I\frac{d^2\Phi}{dt^2} + C(\Phi - \Omega_0 t) = -T^c - T^f \end{aligned} \tag{4}$$

Where *I, M, D, C, K, U, Φ*, Ω_o, V_o, *T* and *W* represent the following: *I* is the effective mass moment of inertia, *M* is the effective mass, *D* is the axial damping, *C* is the torsional stiffness of the drill pipe, *K* is the axial stiffness of the drill pipe, *U* is the axial displacement, Φ is the angular displacement, Ω_o is the top angular displacement, V_o is the top axial displacement, *T* is the Torque on Bit, and *W* is the Weight on Bit, respectively. The superscripts *c* and *f* denote the cutting and frictional components.

A modified lumped parameter model as illustrated in Fig. 9 was proposed by (Kamel and Yigit, 2014) to study a coupled torsional and axial vibrations of vertical drilling system where the estimation of equivalent system parameters was done by employing a Lagrangian approach. it's a remarkable to highlight that this paper was the first attempt to include dynamics of the complete axial drive system and a modified bit/rock interaction model, to explain the cutting and frictional actions at the bit, were used equations 5 hence the rotational and translational motions of the bit are obtained as a result of the overall dynamic behaviour rather than prescribed functions or constants. It illustrated that the unwanted vibrations could be nearly reduced by an appropriate selection of axial and torsional velocities at the optimum operational parameters that could be identified by a suggested optimization method and recommended to extend the analysis using continuous model and to incorporate the effect of lateral vibrations. (Li et al., 2021) agreed to the Kamel and Yigit 2014 results as well and researched on coupled axial and torsional using harmonic balance and alternating frequency/time domain (*HB-AFT*) method and simulated it numerically. Moreover, the formation stiffness effect on drill strings response was studied by them where it was concluded that the increase in formation stiffness increased the response amplitude.

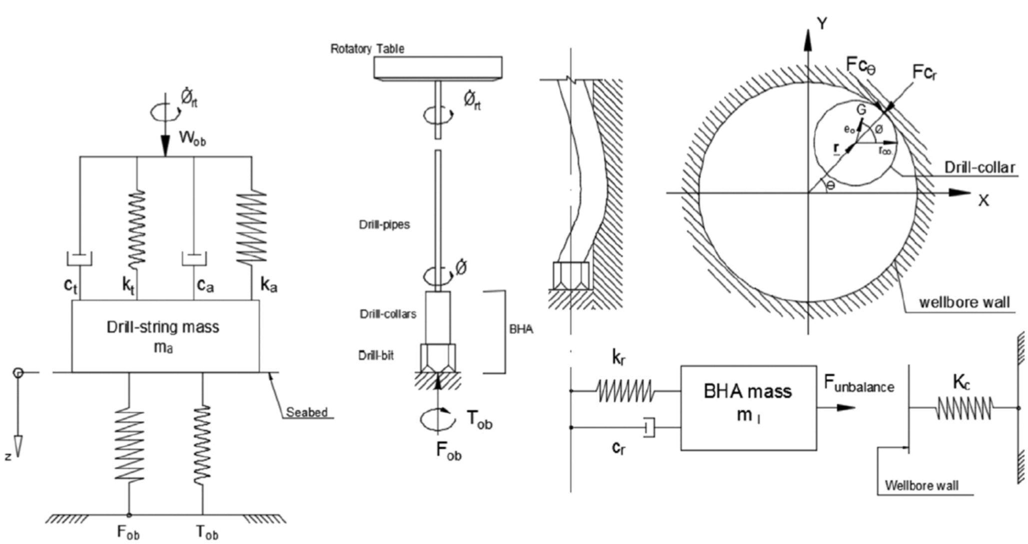

Fig. 10. Drill-string model (de Moraes and Savi, 2019).

$$\begin{aligned}
&\boldsymbol{m_{Top}}\ \ddot{\boldsymbol{x}}_{Top} + \boldsymbol{C_{ds}}(\dot{\boldsymbol{x}}_{Top} - \dot{\boldsymbol{x}}_{a}) + \boldsymbol{k_{ds}}(\boldsymbol{x}_{Top} - \boldsymbol{x}_{a}) = \boldsymbol{m_{Top}}\ \boldsymbol{g} - \boldsymbol{F}_{hook}\\
&\boldsymbol{m_a}\ddot{\boldsymbol{x}}_{a} + \boldsymbol{C_{ds}}(\dot{\boldsymbol{x}}_{a} - \dot{\boldsymbol{x}}_{Top}) + \boldsymbol{k_{ds}}(\boldsymbol{x}_{a} - \boldsymbol{x}_{Top}) = \boldsymbol{m_a g} - \ WOB\\
&\boldsymbol{J}\ddot{\boldsymbol{\theta}} + \boldsymbol{C_V}\dot{\boldsymbol{\theta}} + \boldsymbol{K}(\boldsymbol{\theta} - \boldsymbol{\theta_{rt}}) = \ TOB\\
&(\boldsymbol{J_{rt}} + \boldsymbol{n_o^2 J_m})\ddot{\boldsymbol{\theta}}_{rt} + \boldsymbol{K}(\boldsymbol{\theta_{rt}} - \boldsymbol{\theta}) + \boldsymbol{C_{rt}}\dot{\boldsymbol{\theta}}_{rt} - \boldsymbol{n_o T_{mo}} = 0
\end{aligned} \tag{5}$$

$$\begin{gathered}
m_l\left(\ddot{r} - r\dot{\theta}^2\right) + c_h|V|\dot{r} + k_r r = m_l e_0\left[\dot{\varnothing}^2 cos(\varnothing - \theta) + \ddot{\varnothing} sin(\varnothing - \theta)\right] - F_{cr}\\
m_l\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) + c_h|V|r\dot{\theta} = m_l e_0\left[\dot{\varnothing}^2 sin(\varnothing - \theta) + \ddot{\varnothing} cos(\varnothing - \theta)\right] - F_{c\theta}\\
m_a\ddot{z} + c_a\dot{z} + k_a z = W_{ob} - F_{ob}\\
J\ddot{\varnothing} + c_t\dot{\varnothing} + k_t(\varnothing - \varnothing_{rt}) + c_h|V|\dot{r}_0 sin(\varnothing - \theta) - c_h|V|r\dot{\theta}e_0 cos(\varnothing - \theta) = -T_{ob} - F_{cr}e_0 sin(\varnothing - \theta) + F_{c\theta}\left(\frac{De_{c0}}{2} - e_0 cos(\varnothing - \theta)\right)
\end{gathered} \tag{6}$$

For the axial equations of motion, x_a and x_{Top} represent the axial responses of the bit and the top mass, respectively. The parameters m_a, $m_{Top}, k_{ds}, C_{ds}, g$ and WOB are defined as follows: m_a is the effective mass, m_{Top} is the suspension mass, k_{ds} is the drill string stiffness, C_{ds} is the axial drill string damping, g is the gravitational acceleration, and WOB is the downhole Weight on Bit. For the torsional equations of motion, θ and θ_{rt} represent the angular displacements of the drill bit and the rotary table, respectively. The parameters are defined as follows: J is the lumped inertia of the drill string, J_{rt} is the inertia of the rotary table, J_m is the inertia of the rotary table motor shaft, C_V is the effective damping due to fluid motion around the drill string, K is the effective torsional stiffness of the drill string, C_{rt}, is the torsional internal damping in the gearbox, n_o is the gear ratio of the gearbox of the rotary table motor, T_{mo} is the torque at the motor shaft driving the rotary table, and TOB is the Torque on Bit.

A fully coupled Lumped parameters model was presented as shown in Fig. 10 that taking in account axial, torsional and lateral vibrations by (de Moraes and Savi, 2019) refer to Eq. (6). A comprehensive parametric analysis has been conducted, which encompasses the investigation of various drill string vibration phenomena, including stick-slip, bit--bounce, and whirl. Crucially, this analysis also focuses on the coupling effects arising from the combination of these three vibration modes. The simulations outcomes showed agreements with field data and operational mitigation strategies are proposed to overcome the unwanted vibrations.

The axial, radial, top angular displacement, drill string angular displacement, and geometric center angular displacement are represented by z, r, $\varnothing_{rt}$, $\varnothing$, and θ, respectively. The stiffness of the axial, torsional, and lateral modes are respectively denoted by k_a, k_t, and k_r. The dissipation associated with the lateral mode is denoted by the hydrodynamic coefficient C_h. V represents the velocity of the drill-string geometric center; C_a is the axial dissipation, while c_t is the rotational dissipation. e_0 represents the system eccentricity and De_{c0} is a drill collar Diameter. The weight on the bit and torque on the bit are denoted by W_{ob} and T_{ob}, respectively. $F_{c\theta}$ and F_{cr} represent the lateral reaction forces. m_a, m_l and J represent the drill string mass, BHA mass, and torsional inertia, respectively.

2.2. Distributed model

As far as the authors are aware, (Bailey and Finnie, 1960) and (Finnie and Bailey, 1960) at the Shell Development Company worked together to create the first version of drilling vibration modelling in the 1960s. Their pioneering work introduced a wave equation model specifically designed to analytically address both torsional and axial drill string vibrations. The distributed modelling method demonstrates how this

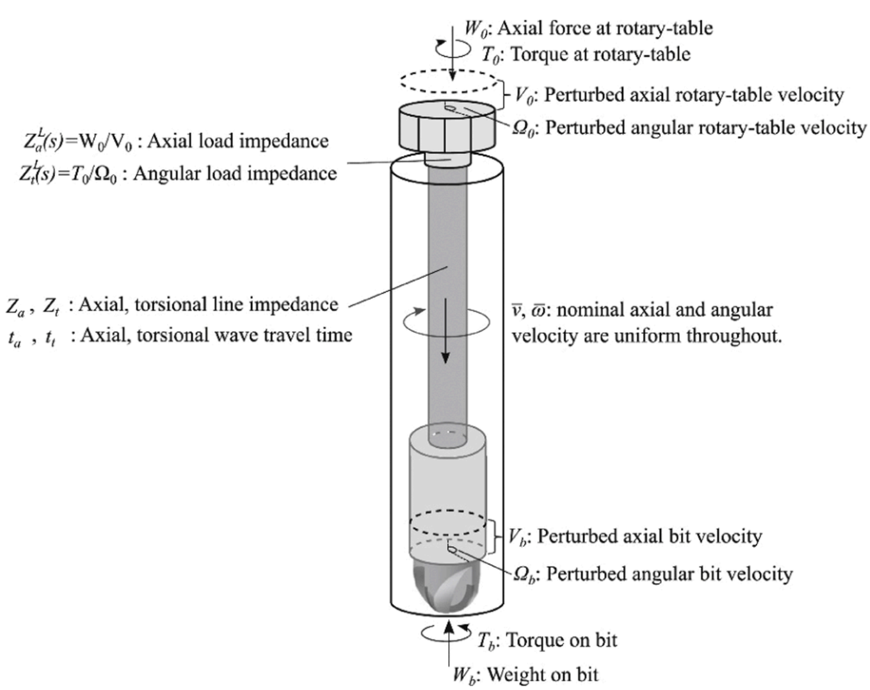

Fig. 11. drill string Schematic (Aarsnes and van de Wouw, 2018).

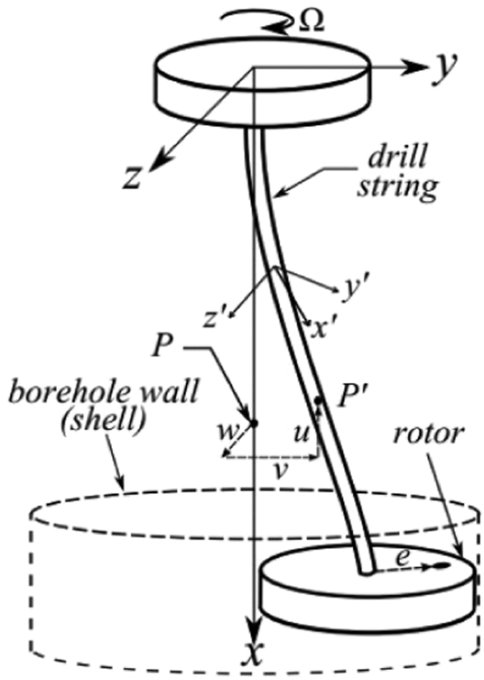

Fig. 12. The drill string model schematic (Vlajic et al., 2012).

model's accuracy in capturing the distributed features of drilling behaviour has greatly impacted a number of recent advances in drilling analysis and control.

Characterizing the linearized torsional and axial interactions within continuous drill string model enabled Aarsnes & van de Wouw, 2018 (Aarsnes and van de Wouw, 2018) to study the drill string dynamics. The research goes beyond the common lumped-parameter models by using both the wave equation approach and a stability analysis in the frequency domain to derive the mathematical description of the drilling system's dynamics, which is schematically depicted in Fig. 11. based on wave equations model that could be obtained by applying the momentum balance acting on an infinitesimal drill string element and continuity equations based on the assumption of elastic deformations. The equation of the axial dynamics is given as (Eq. (7)) below.

$$\begin{aligned} &\frac{\partial w(t,x)}{\partial t} + AE\frac{\partial v(t,x)}{\partial x} = 0 \\ &A\rho\frac{\partial v(t,x)}{\partial t} + \frac{\partial w(t,x)}{\partial x} = -k_a\rho A v(t,x), \end{aligned} \tag{7}$$

Where A refers to the cross-sectional area, whereas $v(t, x)$ and $w(t, x)$ reflect the axial velocity and force, respectively. L represents the length of the drill string. E represents Young's modulus, ρ represents the pipe mass density, and k_a is a damping constant that accounts for shear stresses and structural damping. Similarly, in the context of torsional motion, the quantities representing the rate of change of angular position and the torsional torque are referred to as angular velocity (ω) and torque (τ) respectively. Eq. (8) provides the angular dynamics.

$$\frac{\partial \tau(t,x)}{\partial t} + JG\frac{\partial \omega(t,x)}{\partial x} = 0$$

$$J\rho\frac{\partial \omega(t,x)}{\partial t} + \frac{\partial \tau(t,x)}{\partial x} = -k_t\rho J\omega(t,x) \tag{8}$$

where J is the polar moment for inertia and G is the shear modulus and k_t is a damping constant representing the combined influence of the shear stress and structural damping. The paper's key contributions lie in the novel use of inverse gain margin for stability assessment and the creation of detailed stability maps based on non-dimensional parameters, which underscore the complex of the drill string dynamic, bit rock interaction, and operational conditions. The analysis pinpoints the stabilizing effects of increased rotational speeds and reduced wave reflections at the top drive. The results expose the limitations of lumped-parameter models, particularly in their failure to capture multiple axial and torsional modes that may influence overall system behavior. The findings have important implications for the drilling systems design and control, offering a quantitative framework for addressing vibrations and instabilities through better system parameters understanding. Future work is set to explore the non-linear dynamics using the established modeling framework, which promises to enhance transient simulation models' accuracy further. Study presented by the same authors 2019, Aarsnes and van de Wouw (Aarsnes and van de Wouw, 2019) considered a nonlinear continuous coupled torsional and axial drill string model at bit and a rate independent bit rock interaction law, the torsional and axial

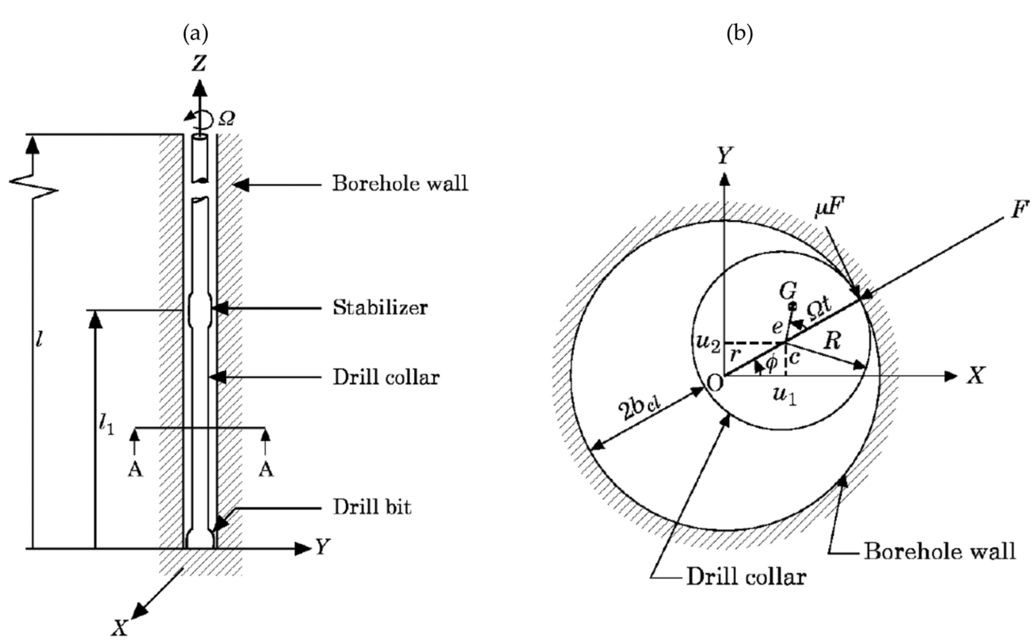

Fig. 13. (a): The system sketch; (b) Borehole wall and deflected drill collar interaction at section A-A (Christoforou and Yigit, 1997).

motion is governed by the partial differential equations. studied the occurrence and the interrelation of the self-excited axial and torsional oscillations that was triggered by the regenerative effect at bit on the bottom hole. Authors performed a stability map for stick slip occurrence and demonstrate that multiple axial modes of the drill-string could be excited or reduced, based on the bit torsional speed and a approve that the complex drilling system is captured sufficiently by distributed model. (Vlajic et al., 2012) formulated a continuous model, as depicted in Fig. 12, to capture the lateral and torsional coupled vibrations considering the interaction between the drilling system and bore wall. The models for tangential and normal contact forces are given in Eq. (9). Both the stick-slip and whirling phenomena were studied numerically and found to be in agreement with experimental results

$$\begin{aligned} F_{normal} &= \begin{cases} 0 & for\ \rho \le \delta \\ K_c(\rho - \delta) & for\ \rho > \delta \end{cases} \\ \mu &= \begin{cases} \left|\dfrac{V_{rel}}{V_{min}}\right|\mu_o & for\ |V_{rel}| < V_{min} \\ \mu_o & for\ |V_{rel}| \ge V_{min} \end{cases} \\ F_{tan} &= -\mu \cdot sign(V_{rel})F_{normal} \end{aligned} \tag{9}$$

Where K_c Contact stiffness between rotor and outer shell, μ Effective friction coefficient, μ_o Friction between lower disk and outer shell, V_{rel} Relative speed between lower disk and outer shell, V_{min} Minimum relative speed for sticking ($V_{min} = 0.02$ m/s), ρ Density of drill string and δ Separation between lower disk and shell.

A coupled distributed model for axial and transverse vibrations of a rotating drill string was developed by (Christoforou and Yigit, 1997). Assuming that the drill string operates at a constant torsional speed. The drill string is modeled as a slender beam with a simply supported lower part for transversely and Fix-Free axially as shown in Fig. 13(a), The Assumed Modes Method and the Lagrangian approach are used to derive the governing equations of motion (Eq. (10)), Full coupling between the transverse and axial vibration modes arises from the formulation's inclusion of nonlinear coupling terms. The model considers potential interactions between the drilling system and the borehole wall, effects of gyroscopic moments, bit-rock interaction, axial excitation, and hydrodynamic damping due to the drilling mud outside the drill string. The drill string and borehole wall contact is captured by using the Hertzian contact theory (Eq. (11)) to model impacts between them, as shown in Figure 13(b). The significance of using a fully coupled analysis, rather than an uncoupled one, was emphasized.

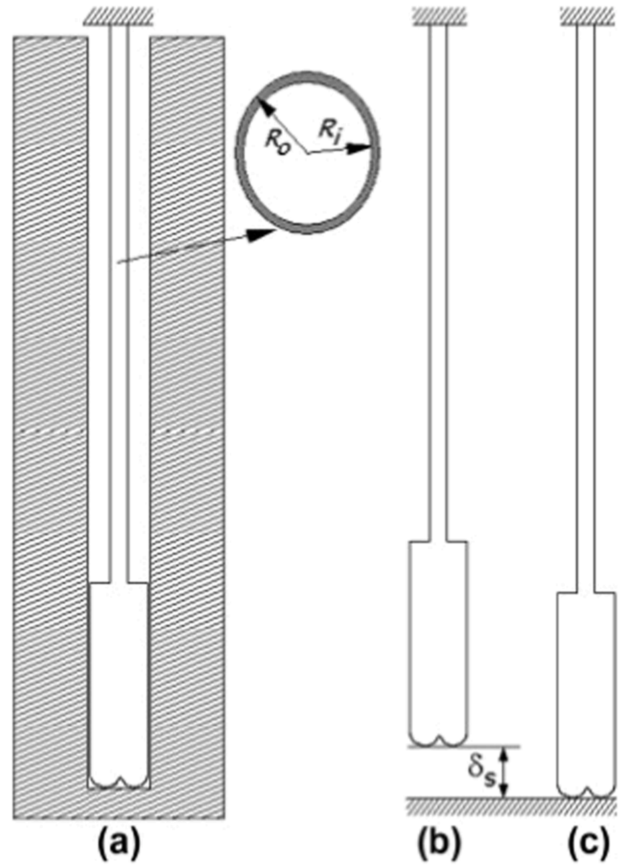

Fig. 14. (a) Drill-string model description; (b) Undeformed and (c) Deformed shape (Sampaio et al., 2007).

$$\begin{aligned} &(1+m_i^*)\ddot{q}_i + \omega_i^2 q_i + \sum_{j=1}^{N}\sum_{k=1}^{R} K_{ijk} q_j \gamma_k - 2G_i\Omega\dot{\eta}_i - S_i e\Omega^2 \cos\Omega t = Q_i,\ i = 1,2,\ldots,M \\ &\left(1+m_j^*\right)\ddot{\eta}_j + \omega_j^2 \eta_j + \sum_{i=1}^{M}\sum_{k=1}^{R} K_{ijk}\eta_i\gamma_k + 2G_j\Omega\dot{q}_j - S_j e\Omega^2 \sin\Omega t = \overline{Q}_j,\ j = 1,2,\ldots,N, \\ &\ddot{\gamma}_k + \varpi_k^2\gamma_k + \frac{1}{2}\sum^{M}\sum^{N} K_{ijk}\left(q_i q_j + \eta_i \eta_j\right) + c_k \ddot{P} = 0,\ k = 1,2,\ldots,R, \end{aligned} \tag{10}$$

where ϖ_k and ω_i are the characteristic frequencies for the axial and transverse the vibrations, respectively. the $q_i(t)$, $\eta_j(t)$ and γ_k (t) are unknown generalized co-ordinates. P is axial load. and $m_i^*, K_{ijk}, G_i, S_i, c_k, Q_i$ and Q_j are obtained from the modal integrals as

$$\begin{aligned} m_i^* &= \int_0^{l_1} \rho I(\phi_i')^2 dz, K_{ijk} = \int_0^{l_1} EA\phi_i'\phi_j'\beta_k' dz \\ G_i &= \int_0^{l_1} \rho I\phi_i\phi_i dz, S_i = \int_0^{l_1} \rho A\phi_i dz, \\ c_k &= \frac{\rho}{E}\left[\int_0^{l_1} \beta_k(z) z dz + \int_{l_1}^{l} \beta_k(z) l_1 dz\right], \end{aligned}$$

The impact force F is obtained from the Hertzian contact law as

$$\boldsymbol{F} = \begin{cases} -\boldsymbol{sign}(\boldsymbol{r_c})\boldsymbol{K_h}(\boldsymbol{r} - \boldsymbol{b_{cl}})^{3/2}, & if\ |\boldsymbol{r_c}| \geqslant \boldsymbol{b_{cl}}, \\ 0, & if\ |\boldsymbol{r_c}| < \boldsymbol{b_{cl}}, \end{cases} \tag{11}$$

where K_h is the Hertzian stiffness which is obtainable from the material characteristics and the contact geometry, r_c is the position of the geometric centre at the impact location and b_{cl} is the borehole clearance.

(Sampaio et al., 2007) delve into the complex dynamics of drill-string

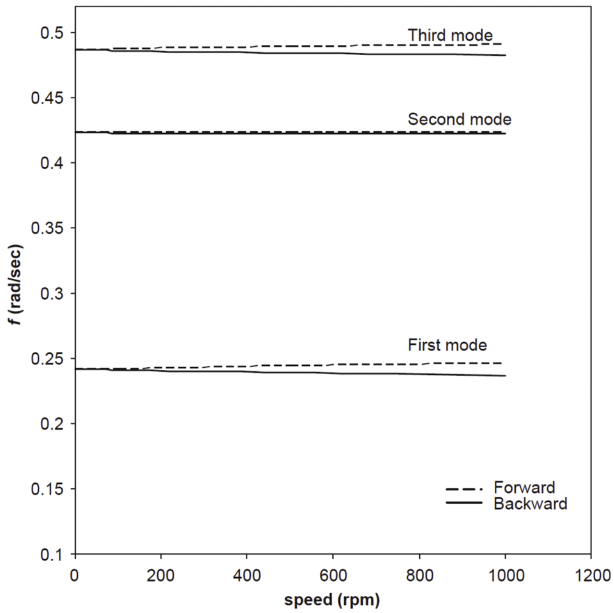

Fig. 15. Bending frequency for neutral point located at the bottom (Khulief and Al-Naser, 2005).

vibrations, emphasizing the pivotal role of geometric non-linearities. They employ a geometrically non-linear model to explore the axial and torsional coupling vibrations in a vertical slender beam under axial oscillation, which represents the behavior of the drill-string, as shown in Fig. 14. The study highlighted that the significance of geometric stiffening, analysed through a non-linear finite element method that incorporates large rotations, structural damping and non-linear strain-displacements.

The finite element method (*FEM*) used is crucial for this analysis where the discretized equation of motion is (Eq. (12)):

$$M\ddot{q} + D\dot{q} + \left[K_e + K_g(q)\right]q = F_g \tag{12}$$

where F_g is the global vector of gravity forces and M, D, K_e, and K_g are the global matrices of mass, damping, elastic stiffness, and geometric stiffness, respectively. The formulated nonlinear *FEM* allows for the consideration of large rotations and non-linear strain-displacements, providing a more accurate representation of the drill-string behaviour. The paper contrasted the non-linear model with its linear counterpart, revealing considerable discrepancies, particularly evident after the initial periods of stick-slip and exacerbated by increased friction at the drill string lower part. The drill-string's behaviour under these conditions is carefully studied, showcasing the model's ability to capture the complex interactions, and highlighting the limitations of linear models in accurately predicting drill string dynamics. The study concludes that geometric non-linearities play a vital role in drill-string vibrations, necessitating the use of non-linear models for a comprehensive understanding and accurate predictions. The influence of these non-linearities is particularly pronounced in stick-slip simulations, underscoring their importance in the analysis of drill string dynamics.

(Khulief and Al-Naser, 2005) created a Finite Element model that encompasses both drill collars and drill pipes. the energy method was used into the variational form of Lagrange equation, and the finite element method technique to derive the govern equation of motion of the drilling system as shown in Eq. (13).

$$[M]\ddot{e} + [G]\dot{e} + [K]e = Q \tag{13}$$

where *{e}* is the deformation vector, M is the system's global assembled mass matrix, G is its gyroscopic matrix, and K is its global assembled stiffness matrix. The gyroscopic matrix G is skew symmetric, whereas the mass matrix M and stiffness matrix K are symmetric. The torsional/bending inertia coupling, the gyroscopic effect and the effect of the gravitational force field were considered in their model. The developed model is integrated into a computational framework for determining modal characteristics, estimating the natural frequencies of the whole drill string, as shown in Fig. 15, and conducting time-response analyses. Notably, the paper underscores the importance of treating the entire drilling assembly as an integrated system, as previous models often focused on the bottom hole assembly or considered drill strings as rotating shafts. The study concludes with an overview of the methodology and its potential for further development to incorporate additional dynamic effects related to wellbore and drill pipe contact, interactions between the drill string and drilling mud flow, and stick-slip at the bottom. and directional drilling.

(Khajiyeva et al., 2024) develop a nonlinear distributed parameter model to analyse axial, lateral, and torsional **vibrations** of the drill-string under seismic load effects, focusing on the impact of seismic

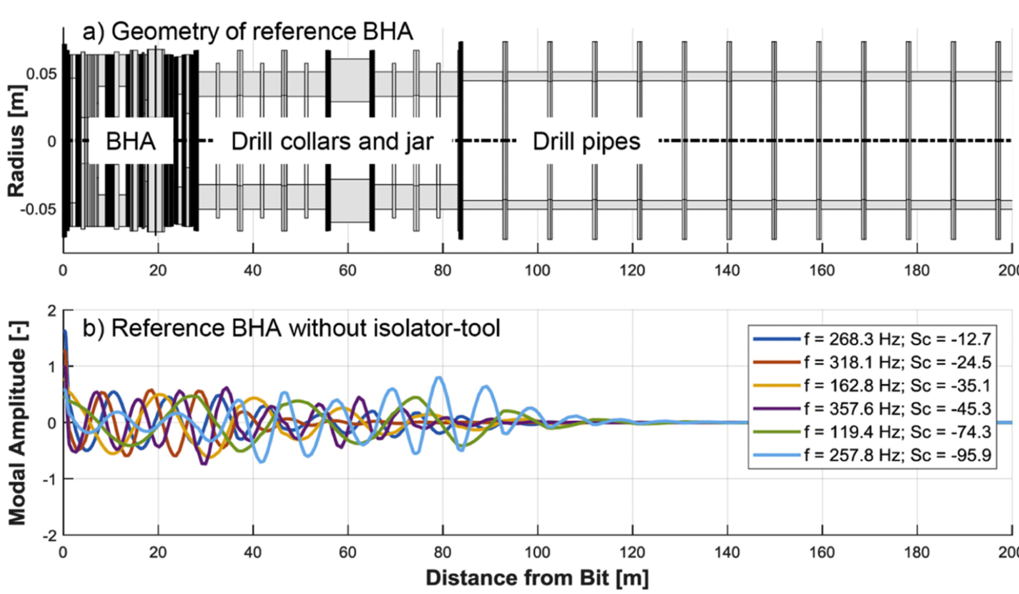

Fig. 16. The top figure represents *BHA* geometry, and the bottom represents the *HFTOs* vibration modes of Bottom Hole Assembly with absence of the isolator tool (Heinisch et al., 2018).

waves on borehole stability. Using the Ostrogradsky-Hamilton variational principle, the model captures spatial deformations, cross-cutting displacements, and angular variations of the drill string, providing a more comprehensive approach than lumped-parameter models. The study employs nonlinear elasticity theory to incorporate higher-order strain tensors, highlighting the limitations of linear models in predicting seismic-induced vibrations. Numerical simulations using Wolfram Mathematica reveal significant amplification of vibrations, particularly in longitudinal and torsional modes, as seismic activity increases. The research identifies beat phenomena, caused by seismic waves, which can destabilize drill-string motion, leading to potential failures in seismically active drilling regions. The findings emphasize the importance of nonlinear modelling in designing robust drilling strategies, ensuring stability and operational safety in seismic environments.

(Faghihi et al., 2024) develop a distributed mathematical model to analyse axial-torsional vibrations in drill-strings based on wave propagation principles, particularly focusing on stick-slip and bit-bounce effects, using Delay Differential Algebraic Equations (DDAEs), the model captures multiple regenerative effects in bit-rock interaction, improving upon traditional Partial Differential Equation (PDE) approaches. A novel depth-of-cut calculation method is introduced, reducing computational complexity while maintaining accuracy in capturing state-dependent delays that drive unwanted oscillations. The study provides stability maps and eigenvalue analyses to identify conditions leading to self-excited vibrations, offering insights into safe drilling parameters. Numerical simulations confirm how delay effects influence system stability, highlighting critical areas for performance optimization and the effects of including the top drive dynamic on simulation accuracy. These findings aid in designing vibration suppression strategies, ultimately enhancing drilling efficiency and reducing operational costs. (Faghihi et al., 2024) propose an active control strategy to suppress axial-torsional vibrations in drill-strings using a Neutral-type Delay Differential Equation (NDDE) model. A wave reflection compensator eliminates neutral delay terms, simplifying the system for a state-feedback controller that ensures stability with an exponential decay rate. A prediction-based control approach using top-side measurements makes the system practical for real-world applications. Simulation results confirm the controller's ability to suppress stick-slip and bit-bounce, demonstrating robustness against uncertainties and noise. Unlike previous models, this parametric control framework adapts to various drilling conditions, reducing vibration-induced failures and improving operational efficiency.

3. High frequency oscillations

High-frequency torsional oscillations (*HFTOs*) occur predominantly in hard and dense rock formations due to the bit-rock cutting action and high operational speed. These oscillations are associated with self-excitation of higher torsional modes, typically ranging from frequencies above 50 Hz to around 400 Hz. Such excitation is notably observed with polycrystalline diamond compact (*PDC*) cutters engaging with these hard formations, as highlighted by (Jain et al., 2014). The *HFTOs* detected frequencies match exactly with natural frequencies of the torsional vibration mode of the drilling system (Jain et al., 2014; Oueslati et al., 2013). *HFTO* can be differentiated by vibrations mode shapes that are trapped within the *BHA* (Fig. 16) and natural frequencies that are typically above 50 Hz. Significant challenges with *HFTOs* are the localized to the *BHA* and it often undetectable nature by surface sensors, leading to an illusion of seamless drilling until unforeseen downhole tool failures or *BHA* components damage occurred. This makes the modelling of High Frequency Torsional Oscillations crucial for maintaining optimal drilling system performance. *HFTO* modelling primarily focuses particularly on the *BHA* rather than the entire drill string, as these vibrations are typically localized in the *BHA*.

(Ichaoui et al., 2020) developed an augmented Kalman filter in 2020 by combining the finite element model of the *BHA* with data acquired from the real drilling system's sensors spread throughout various Bottom Hole Assembly (*BHA*) components that can simulate the torsional vibrations for low and high-frequency torsional oscillations (*HFTOs*). With an approximation model, the input and output estimate technique described in this study can conduct good *BHA* load detection. The suggested technique was appropriate for monitoring Bottom Hole Assembly (*BHA*) vibrations during drilling utilizing limited data on real-time load or creating a digital twin for the Bottom Hole Assembly.

(Chen et al., 2019) created a model using the three-dimensional finite element method to replicate the transient dynamic behaviour the high-frequency torsional oscillations (*HFTO*). The model was specifically designed to study the effects of using a rotary steerable system or downhole mud motor tools. It also investigated the impact of changes in the design of the Bottom Hole Assembly (*BHA*) and drilling parameters on the severity of HFTO. The model incorporated the bit-rock interaction by utilizing cutting forces obtained from single cutter testing for several orientation angles, different sizes of the bit cutters and depth of cut with various type of formations. The architecture of the mud motor, both external and internal, was also modelled. with unique components to meet the motor power criteria. Furthermore, the simulation of the model was utilized to replicate the transient dynamic behaviour of a field-operated motorized Rotary Steerable System (*RSS*) Bottom Hole Assembly (*BHA*). This simulation was then compared with high-frequency bottom hole information measurements to validate its performance. the response of the drilling system was described by following equation (Eq. (14)):

$$[M]\cdot\ddot{U} + [C]\cdot\dot{U} + [K]\cdot U = F(t)) \tag{14}$$

The symbols *M, C,* and *K* indicate the assembled mass, damping matrices, and stiffness of the drilling system, respectively. U, $\dot{U}$and $\ddot{U}$ represent the displacement, velocity, and acceleration vectors, respectively. *F(t)* represents the assembled force vector that acts on the system. Eq. (15) was used to calculate the natural frequencies of the torsional mode of the Bottom Hole Assembly assuming free and free boundary conditions.

$$\omega_n = \frac{n\pi}{L}\sqrt{\frac{G}{\rho}}, (n=1,2,3...) \tag{15}$$

where ρ is the density of the material, L indicates the drill collar length, n is the number of the vibration mode and G represents the modulus of the shear. The model has been validated by a notable degree of consistency between the simulation results and the field data. In order to capture *HFTOs*, (Tikhonov and Bukashkina, 2014) produced a dynamic three-dimensional model drill string. The drill string, bottom hole assembly (*BHA*), positive displacement motor (*PDM*), drilling bit, and formation properties were all taken into consideration by this model. The bit-rock interaction allowed the model to consider the drilling system's torsional and axial coupling vibrations. Additionally, simulated mud motor excitations were used to look at *HFTOs* during slide and rotary drilling. The following (Eq. (16)) was used to express the

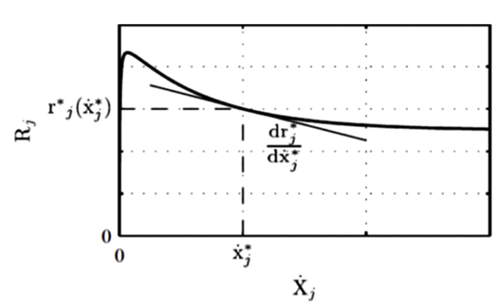

Fig. 17. the characteristic of the falling friction/torque of the contact force/torque in relation to the bit torsional velocity (Hohl et al., 2015).

Table 1
Summary of influential mathematical models that have significantly advanced drill string vibration analysis.

No.	Author(s)	Frequency Range	Model Type	Model Description	Improvements Provided
1	(Belokobyl'skii and Prokopov, 1982)	Low	Discrete (Lumped)	Developed a 1 DOF torsional pendulum model to analyze self-excited torsional vibrations	Laid foundational work for understanding torsional vibrations but did not include mud damping effects
2	(Yigit and Christoforou, 1998)	Low	Discrete (Lumped)	Developed a nonlinear model for torsional and bending vibrations	Demonstrated energy transfer between lateral and torsional modes
3	(Elsayed et al., 1997)	Low	Discrete (Lumped)	Studied multi-DOF axial and torsional vibrations	Identified critical stability conditions for stick-slip
4	Germay and Detournay (2004–2007) (Richard et al., 2004; T. Richard et al., 2007)	Low	Discrete (Lumped)	Investigated self-excited stick-slip vibrations using a PDC bit and coupled torsional and longitudinal modes	Introduced a nonlinear model considering bit-rock interaction and cutting forces for predicting stick-slip
5	(Rudat and Dashevskiy, 2011)	High	Discrete (Lumped)	Modeled torsional pendulum for stick-slip phenomena in drilling	Proposed optimization of drilling parameters to reduce stick-slip
6	(Tang and Zhu, 2020)	Low	Discrete (Lumped)	Investigated the impact of drill string length on the existence of stick-slip oscillations	Demonstrated that increased drill string length exacerbates torsional amplitude and slip phase duration
7	(Kamel and Yigit, 2014)	High	Discrete (Lumped)	Coupled axial and torsional vibration model	Proposed a method to optimize axial and torsional velocities
8	(Sampaio et al., 2007)	High	FEM	Developed a nonlinear finite element model for axial and torsional coupling in vertical slender beams	Highlighted the importance of geometric stiffening and nonlinear strain displacements for accurate modeling
9	(Chen et al., 2019)	High	FEM	Modeled high-frequency torsional oscillations (HFTO) in a 3D FEM model	Improved Rotary Steerable System (RSS) modeling, validating HFTO impacts on bit design
10	(Tikhonov and Bukashkina, 2014)	High	FEM	Developed a 3D model of drill string torsional and axial vibrations including positive displacement motor (PDM)	Simulated HFTO during slide and rotary drilling, showing the impact of PDM dynamics
11	(Aarsnes and van de Wouw, 2019)	High	Distributed	Studied torsional and axial dynamics using wave equation model, focusing on stick-slip and multiple axial modes	Developed stability maps, showing complex interrelations between torsional speed and axial modes
12	(Bailey and Finnie, 1960; Finnie and Bailey, 1960)	High	Distributed	Developed the first wave equation model to capture both torsional and axial drill string vibrations	Pioneered distributed modeling, greatly impacting drilling vibration analysis and control
13	(Besselink et al., 2011)	High	Distributed	Developed a continuous model to capture lateral and torsional vibrations, considering drill system and borehole interactions	Proposed a method to stabilize torsional vibrations by stabilizing axial ones using continuous coupling
14	Vlajic et al. (2011) (Vlajic et al., 2012)	High	Distributed	Modeled lateral and torsional coupled vibrations considering drill system and borehole interaction	Studied stick-slip and whirling phenomena numerically, with experimental agreement
15	(Hohl et al., 2015)	High	FEM	Developed a stability criterion to determine critical torsional modes that induce HFTOs	Validated the model with field data to predict HFTO frequency, showing that higher modes are prone to HFTOs
16	(X. Li et al., 2021)	High	Discrete (Lumped)	Researched coupled axial and torsional vibrations using harmonic balance and time domain methods	Found that increased formation stiffness raises the response amplitude, affecting drill string behavior
17	(M.A. Faghihi et al., 2024)	Low	Distributed (DDAE & NDDIE)	Developed a distributed drill-string model incorporating axial-torsional vibrations with multiple regenerative effects in the bit-rock interaction	Introduced a new depth-of-cut calculation method, preserved infinite-dimensional nature, provided stability maps, and improved computational efficiency
18	(M.A. Faghihi et al., 2024)	Low	Distributed (NDDE with Prediction-Based Control)	Developed a control strategy for suppressing axial-torsional vibrations using state-feedback and prediction-based control to mitigate wave reflections and input delays	Introduced a wave reflection compensator, ensured exponential stability, employed a prediction-based feedback controller, and enhanced robustness against uncertainties and noise

dynamics of axial and torsional vibrations:

$$\begin{aligned} m\frac{\partial^2 U}{\partial t^2} - EA\frac{\partial^2 U}{\partial s^2} &= q \\ \rho I_p\frac{\partial^2 \Phi}{\partial t^2} - GI_p\frac{\partial^2 \Phi}{\partial s^2} &= p \end{aligned} \tag{16}$$

Where U indicates the displacement in axial direction. Φ is the angle of rotation. ρ represents the density of the material. m represents the mass per length. The symbol E represents Young's modulus, while the symbol G indicates shear's modulus. The variable A represents the area of the cross-section of the Bottom Hole Assembly (*BHA*). Ip represents the polar moment of inertia. q denotes the external applied force per length, while p indicates the external torque per length. The variable s represents the position of the arc, measured from the bottom of the drill string along its axis. t is the variable of time.

It's found that the bit was able to generate *HFTOs* with significant amplitudes on its own, according to the Authors, because of the positive displacement motor (*PDM*), which prevented a stiff connection between the bit and the remainder of the *BHA*. Furthermore, the dynamics of the positive displacement motor (*PDM*) was proposed to be a major determinant of the *HFTO* frequencies. (Abedrabbo. and Lines., 2015) employed finite element methods (*FEM*) to analyse a complicated drilling assembly both statically and dynamically inside a well-bore using commercial software. That utilized in the drill string representation to reproduce the vibrations that are coupled. The Wellbore contact frictions, curvature of the well, the gravity effects, drill pipes eccentric, and *BHA* components operation all contributed to complex non-linear boundary conditions that considered in simulation in the frequency and Time domains. A high frequency peaks seen in tangential acceleration during the drilling of chalk and marl formations were matched with the predicted *HFTOs* by simulation at the dominating frequency. Additionally, the model clearly displayed destructive whirling modes and lateral vibrations. According to this study, the finite element approach is an invaluable tool for accurately and efficiently analysing the dynamic response of a drill string with intricate three-dimensional shapes and structures. Predicting the dynamic behaviours of drilling system in different configurations is made possible by the method's ability to include genuine *3D* components into the analysis, which allows for the modelling of real physics. A stability criterion was suggested by (Hohl et al., 2015) in 2015 to determine the critical torsional modes that have a higher probability of inducing *HFTOs* as shown in Fig. 17. The assumption used to derive this analytical criterion was that the downhole cutting process would have the characteristics of the velocity-weakening only excited one natural mode of torsional oscillations, i.e., At a constant Weight On Bit, the torque/friction generated from the rock breaking process reduces as the drill bit's torsional velocity increases

The criteria considered the interaction between the contact forces weakening ratio and the BHA's modal properties as determined by finite element analysis. that was expressed in Eq. (17) below:

$$S_{c,i} = -\frac{2D_i\omega_{0,i}}{\varphi_{ii}^2} < \frac{dr_j^*}{dx_i^*} \tag{17}$$

The value of $S_{c,i}$ represents the minimum slope of the torque characteristic at which the system is just stable. $\omega_{o,i}$ and D_i represent natural frequency and the modal damping, respectively, of the i th mode. The variable φ_{ij} represents the deflection of the j-th node in the i-the mode shape. The term; $\frac{dr_j^*}{dx_i^*}$ refers to the slope of the contact forces at the j-th node in the finite element model. It is important to observe that $S_{c,i}$ varies for each mode i, and that $S_{c,i}$ is characterized by a negative value, which implies that higher modes levels are prone to *HFTOs*. The authors' field case study, in which the frequencies of the critical torsional modes were detected between 54 Hz and 351 Hz, served as validation for this criterion. The measured mode frequencies and shapes and those computed using the finite element model agreed well.

Table 1 provides a structured summary of key mathematical models addressing drill string vibrations. It lists the authors**,** frequency range**,** model type**,** model description**,** and improvements provided in each study. The table highlights fundamental advancements in vibration modelling, including nonlinear coupling, stability analysis for low and high frequencies, which have contributed to a better understanding of drill string dynamics. These studies offer crucial insights into the evolution of mathematical modelling approaches and their impact on drilling system performance.

4. Experiment methods

Experimental methods are crucial for understanding drill-string vibrations and optimizing drilling operations. Advancements from basic axial-lateral setups to multi-degree-of-freedom and autonomous systems have enhanced accuracy and applicability. These models validate simulations, improve efficiency, and guide vibration mitigation strategies, playing a key role in real-world drilling advancements.

4.1. Examples of the experimental models

Experimental models play a crucial role in understanding drill-string vibrations, enhance the bit and drill string design and optimize the operation parameters. Various setups have been developed, each offering unique advantages and limitations. Early test rigs primarily focused on axial and lateral vibrations using vertically mounted drill-string models with rigid frame structures, such as the one used by (Antunes et al., 1992; Berlioz et al., 1996) showing that fluid forces and axial excitations influence frequency, damping, and instability behaviour of the drill-strings however they are limited in their ability to simulate axial and lateral vibrations. More advanced test rigs, such as those utilizing lubricated disc brake mechanisms to represent friction forces proposed by (Mihajlovic et al., 2006), that improved torsional vibration control but did not fully capture the axial and lateral vibrations as it is in real drilling environments. These studies are examples of experiments that effectively identified basic instability modes and vibrations behavior, but they lacked the complexity needed to simulate realistic rock-bit interactions. Meanwhile, setups integrating actual bit/rock interactions, (Bavadiya et al., 2017) have provided more realistic drilling data, primarily focusing on the lateral and axial vibrations in a drill-string test rig. The study shows that the axial vibrations were found to be increased when the rotational speed approached the drill string's natural frequency, leading to an increase in the rate of penetration (ROP). The type of rock formation influenced the amplitude of both lateral and axial vibrations. reducing rotational speed and weight on the bit enhances decrease lateral vibrations in hard formations, while lowering rotational speed alone can effectively reduce axial vibrations. However, the study explicitly states that torsional oscillations were not observed due to the stiffness and small size of the drill-string, in addition to the cost of experimental repeatability due to rock heterogeneity and covering axial and lateral vibrations. Multi-degree-of-freedom modified test rigs proposed by (Westermann et al., 2015), introduced a new scaled laboratory rig for analysing drill string vibrations, focusing on lateral and torsional oscillations, covering stick/slip and whirl phenomena. The test rig uniquely applies mechanical similarity laws to ensure experimental results are comparable to real drilling conditions. Unlike existing rigs, it can measure contact forces and allows the study of coupled torsional and lateral vibrations. Experimental results revealed key insights, such as plastic deformation effects on lateral vibrations and the instability of backward whirl at critical speeds, which has implications for drilling operations. However, there are limitations such as, the absence of fluid interaction, and the focus on a single section of the drill string. Additionally, studies on axial impact mechanisms and stick-slip phenomena have been studied by (Wang et al., 2021) they investigated the effect of an axial impactor in suppressing stick-slip vibrations, aiming to confirm field observations regarding its effectiveness. The experimental study verifies that axial impact frequency plays a crucial role in eliminating stick-slip vibrations. Additionally, the results show that the axial impactor reduces fluctuations in weight on bit (WOB) and torque on bit (TOB), further contributing to more stable and efficient drilling, However, there are limitations such as coupling effects with other vibrations modes is not considered, simplified bit/rock interactions, and fixed boundary conditions. Recent research has also demonstrated that autonomous drilling control systems can mitigate drilling vibrations conducted by (Zarate-Losoya et al., 2018) . The study introduces an autonomous lab-scale PDC bit drilling rig designed to enhance efficiency and reduce downhole dysfunctions using machine learning and real-time automation. Main features include a machine learning-based lithology prediction model, an MSE-driven optimization algorithm, and an advanced control system for dysfunction suppression. The system considerably improved drilling efficiency but faced limitations, mainly in lithology prediction due to limited training data, sensor data loss at high RPM, and bit design constraints. Despite these challenges, the research highlights the transformative potential of automation and machine learning in drilling operations. (Kapitaniak et al., 2018)have provided deeper insights into optimizing drilling parameters using real PDC bit and rock sample in their experiment. they studied the drill string whirls, confirming the co-existence of forward and backward whirls, with chaotic motion at low speed/WOB and periodic whirls at higher levels. It highlighted BHA sensitivity to parameter changes and introduced new features like real drill bits and rock samples, coupled vibration analysis, and whirl transitions. However, limitations in scaling, material testing, and controlled environments. The discussed experiments results reinforce the importance of experimental setups in deeply understanding the critical vibrations phenomena, representing downhole dynamics accurately and guiding practical mitigation strategies.

4.2. Experimental models improvement

The evolution of experimental models for studying drill string vibrations has been significantly influenced by early foundational work. (Shyu, 1989) pioneered the study of axial and lateral vibrations using a simple motor-driven setup, axial shaker representing the bit/rock interactions and Strain Gauges, which provided the basis for more advanced models. (Berlioz et al., 1996) expanded on this by incorporating fluid-structure interactions, higher motor velocity, borehole curvature, and stabilizers. It provides more realistic testing conditions to the experimental setup. This progression continued with (Lin et al., 2018), who developed a 25-meter horizontal setup capable of studying all three vibration modes (axial, lateral, and torsional), integrated real-time wireless data acquisition, scaling similarity principles building on the limitations of earlier vertical and inclined setups. These advancements demonstrate how early models have informed and improved later experimental designs, leading to more comprehensive and accurate simulations of drill string dynamics. (Miyazaki et al., 2019) further advanced this field by focusing on the wear of PDC cutters in hard rock formations, demonstrating that the rate of increase of Wear Flat Length is closely related to the indexes of rock abrasivity, providing a quantitative measure of cutter wear that can be used to predict bit life in real-world drilling scenarios using vertical drill setup,Hydraulic Unit, Displacement Transducer, PDC drill bit and rock samples. Continuing In the same objective, (Mao et al., 2024)investigated the dynamic response of a PDC bit, focusing on vibration patterns, cutting efficiency, and variations in weight-on-bit (WOB) and torque-on-bit (TOB). The study utilizes a commercial PDC bit and rock samples to examine how different rock formations influence bit vibrations and cutting depth. The experimental setup includes a hydraulic system for pressure control, an upper turntable for bit rotation, and a lower turntable for rock rotation, enhancing the realism and accuracy of bit-rock interaction simulation. The evolution of laboratory-scale experimental rigs demonstrates a clear trend toward greater realism and control aligning with improvement in drilling rig technology, with modern setups offering improved downscaling methodologies, real-time control capabilities, and enhanced data accuracy.

4.3. Application of experimental models in actual engineering

Experimental models have been instrumental in translating theoretical simulations into real-world drilling optimizations. One of the most significant recent contributions of laboratory-scale setups is their ability to test and validate drilling automation technologies (Zarate-Losoya et al., 2018). A key advancement in mitigating drilling vibrations has been the development and implementation of Soft Torque Rotary Systems (STRS), which actively dampen torsional vibrations at the surface to reduce stick-slip effects (Runia et al., 2013). The application of STRS has demonstrated improved drilling stability in various field applications, as seen in large-scale deployments by Shell and Gulf Drilling International (Attar et al., 2014). Additionally, a new stick-slip prevention system developed by (Kyllingstad and Nessjøen, 2009) has been validated through field testing and Hardware-In-the-Loop (HIL) simulations. Unlike conventional active damping systems that rely on torque feedback, or motor current, this system utilizes a specially tuned PI speed controller to suppress stick-slip vibrations by automatically adapting to the observed oscillation frequency. The system has demonstrated improved efficiency in reducing cyclic torque variations and enhancing drill bit stability in various operational environments. Another critical engineering application is the development of anti-stick-slip drill bit designs. Experimental setups have been used extensively to optimize depth-of-cut (DOC) control technology in polycrystalline diamond compact (PDC) bits, reducing the likelihood of stick-slip and improving bit durability (Schwefe et al., 2014). Filed evidence of DOC control mechanisms on different types of formations has validated their efficiency, showing significant reductions in torque oscillations and overall improvements in drilling speed and tool lifespan. These studies reinforce the importance of experimental validation in improving drilling efficiency and ensuring the reliability of new control methodologies.

5. Boundary conditions

Because of the complex nature of the dynamics of the drill string, particularly the coupling vibration effects, finding exact solutions to the governing equations is nearly impossible. Instead, numerical methods are employed to approximate these solutions. The boundary conditions representation of drill string is at utmost important in order to precisely capture the drill string dynamics during drilling hence an accurate assumptions and acceptable simplifications for the boundary conditions on various modes are a crucial step for obtaining a precise approximation and, consequently, an accurate solution.

5.1. Axial boundary conditions

There are numerous treatment approaches presented in the literature for axial drill string boundary conditions (*BCs*). Common types include fixed at the surface and free at the lower part Boundary Condition, as outlined by (Yigit and Christoforou, 1996), fixed at the surface and fixed at the lower part Boundary Condition described by (Dareing and Livesay, 1968), and free at surface and fixed at the lower part axial Boundary Condition, followed by (Finnie and Bailey, 1960; Paslay and Bogy, 1963). However, (Jogi et al., 2002) argued that the free-fixed axial Boundary Condition doesn't agree well with the recorded axial frequencies in actual system. They posited that assuming a free end provides predictions that align more closely with field observed data. (Aarrestad and Kyllingstad, 1993)examined the top boundary condition, proposing a simple model to represent the top suspension system's dynamics as an equivalent mass spring damper. They provided a method to measure these values and emphasized the need to consider the interaction between transversal drill-lines vibrations and axial drill string vibrations.

They proposed a model that is coupled and nonlinear axially, aiming to evaluate the effects of the top *BCs* on the drill string behaviour. At the bottom boundary, bit displacement, influenced by drilled rock properties and type of the drilling bit, were proposed as a crucial factor needs to be considered for accurate modelling especially in the axial direction. (Spanos et al., 1995) observed that the rolling of the bit creates a multi-lobed surface on the formation, with the number of lobes correlating to the bit's cone data. They studied the influence of this lobed pattern on *WOB* fluctuation and found it could be characterized by a sinusoidal angular variation in its elevation. (Christoforou and Yigit, 1997) described the axial bit displacement fluctuation in a similar sinusoidal profile for the lower Boundary Condition in axial motion for both Tricone and *PDC* bit types. However, they assumed the displacement frequency for tri-cone bits to be triple times the drill string's rotary speed and identical to the torsional velocity for *PDC* bits. (Yigit and Christoforou, 1998; Yigit and Christoforou, 2000) considered a variation in the *WOB* component with the same frequency as the drill string torsional velocity for *PDC* bits. (Besselink et al., 2011) developed an advanced displacement model for accurately measuring the instantaneous depth of cut. This measurement technique relies on assessing the difference between the cutter's axial position on the rock surface at the current moment and its position at a prior time point, denoted as t_n. This model is pivotal in determining the cutting components of the Torque on Bit (*TOB*) and Weight on Bit (*WOB*), that are fundamentally linked to the measured depth of cut. Consequently, this relationship facilitates the coupling of both axial and torsional dynamics within the drill string, through an innovative bit rock interaction law posited by the researchers. Such an approach provides a significantly enhanced and more realistic depiction of the drill string's operational dynamics. Commonly proposed boundary conditions for axial vibration of the drill string in the literature include being fixed at both ends, fixed at the surface and free at the lower part, and a combination of a mass-spring-damper system at the top with either a sinusoidal displacement or an instantaneous depth of cut displacement at the bit location. The last configuration is considered the most realistic because it reflects the system's dynamics instead of relying on a predetermined value.

5.2. Torsional boundary conditions

In the top section of the drill-string, the rotary table is typically represented as a large inertia component operating at a constant rotational speed. Numerous studies, including (Richard et al., 2007; Jansen and van den Steen, 1995; Serrarens et al., 1998; Leine et al., 2002), have modelled the rotary table as a fixed-speed boundary, ensuring stable energy transfer to the drill-string. Additionally, other studies, like Kamel and Yigit (2014) (Kamel and Yigit, 2014), have focused on regulating the speed of the top rotary table through the applied motor voltage however in some cases, control mechanisms are incorporated at the rotary table to regulate the torque-speed relationship, as suggested by (Brett, 1992). This approach aims to mitigate torsional vibrations and maintain operational stability. At the bottom section, the torsional boundary condition is governed by bit-rock interactions, which significantly influence stick-slip oscillations and energy dissipation. One widely used model is the velocity weakening law, where frictional resistance decreases with rotational speed, leading to self-excited vibrations (de Moraes and Savi, 2019; Sharma et al., 2025). Alternatively, the rate-independent model separates frictional losses and cutting forces, showing that torsional energy dissipation depends on the depth of cut and rock properties (Detournay and Defourny, 1992). Recent approaches use Finite Element (FEM) and Discrete Element Models (DEM) for more accurate bit-rock interaction simulations. (Jaime et al., 2015) applied FEM-based erosion modeling to capture plastic and brittle failure, while (Zhang et al., 2022) used DEM to simulate the transition between ductile and brittle rock failure. These advancements improve the representation of the bottom torsional boundary condition, leading to better torsional vibration predictions and drill-string stability analysis.

5.3. Lateral boundary conditions

Various studies have incorporated different lateral boundary conditions (BCs) to analyse the lateral vibrations of drill strings at stabilizer locations. (Choe et al., 2023) utilized simply supported boundary conditions in their study on the impact of torsional stick-slip vibrations on whirl behaviour, providing insights into lateral vibration effects. Similarly, (Popp et al., 2018) employed simply supported BCs while investigating borehole contact models for predicting backward whirl and improving drilling tool reliability. (Dareing, 1984), also applied simply supported conditions when developing control guidelines for drill-string vibrations, further reinforcing their relevance in drilling dynamics. Additionally, (Yigit and Christoforou, 1998; Yigit and Christoforou, 1996) incorporated simply supported boundary conditions in their coupled torsional-bending vibration models to evaluate drill-string interactions with borehole walls, highlighting their influence on overall system stability. Additionally, (Christoforou and Yigit, 2003) modelled the stabilized section of the drill collars as a simply supported beam, assuming the drill-string to be fixed at the top and free at the bit for axial and torsional motion. (Kudaibergenov et al., 2025) examined lateral vibrations of a rotating isotropic elastic rod, where simply supported boundary conditions were imposed at both ends, ensuring zero transverse displacement and moment constraints. The practical applicability of simply supported BCs at stabilizer locations was further validated by (Jogi et al., 2002), where field experiments confirmed model-derived natural frequencies of drill strings. Beyond simply supported conditions, researchers have explored alternative boundary conditions to enhance model accuracy. (Bailey et al., 2008) adopted a free-free boundary condition, capturing scenarios where the drill-string is unconstrained at both ends. Conversely, (Hakimi and Moradi, 2009) utilized a fixed-fixed boundary condition, modelling cases where both ends of the drill-string are rigidly constrained. (Al-Hiddabi et al., 2003) and Zhao and Lian (2008) (ZHAO and LIANG, 2008) introduced deformed springs as lateral boundary conditions, providing a flexible constraint model that simulates stabilizer deflections. (Berlioz et al., 1996) explored the mass-loaded boundary condition, demonstrating how additional mass elements influence lateral vibration behaviour and system response. Several additional studies have applied unique lateral boundary conditions in drill-string models. (Christoforou and Yigit, 1997) considered a single stabilizer positioned at a distance from the bit, where the bit was approximated as simply supported, while the stabilizer imposed a condition between simply supported and fixed. (Khajiyeva et al., 2024) incorporated hinged support at both ends of the drill-string, applying lateral displacement constraints using nonlinear differential equations, influenced by soil motion laws to describe seismic interactions. Regarding top boundary conditions for lateral motion, many studies assume the drill-string is fixed at the surface while allowing lateral movement at the bit. This assumption has been widely applied in the works of (Al-Hiddabi et al., 2003), (Gulyaev et al., 2006), and (Bailey et al., 2008), where they examined stability and equilibrium conditions of rotating drill-strings under various operational constraints. These different boundary condition models contribute to a more comprehensive understanding of drill-string behaviour, aiding in the optimization of stability and performance in downhole operations.

6. Discussion and conclusion

The mathematical modeling of drilling systems is a crucial and complex task. It is a valuable tool used to improve the efficiency and safety of the drilling process, and importantly, to evaluate the influence of changing drilling operation parameters on drilling performance by accurately predicting the behavior of the drilling system.

Including essential components of the drilling system in the mathematical model is of utmost importance for model accuracy. The rotary drilling system consists of several principal components, such as the suspension mass, the draw works system, the rotary table, and the drill string's axial, lateral, and torsional characteristics. Additionally, considering the effects of mud, bit/rock interactions, and drill collar/downhole wall interactions enhances the model's realism and ensures a more accurate representation of the drilling system's dynamics.

Developing an accurate mathematical model of a drilling system, combined with Measurement While Drilling (MWD) tool data, provides numerous benefits. These include the ability to precisely predict system behavior, diagnose encountered faults, propose appropriate solutions, understand the effects of components and operation parameters, optimize the drilling process, and identify new research areas.

In the Lumped Parameter modeling approach, it is observed that the lumped approximation effectively captures the first resonance mode of a drill string at low frequencies. In this model, the drill string is represented as a mass-spring-damper system, described by a set of ordinary differential equations. Significant drilling system characteristics are incorporated, including geometric data, equivalent mass and inertia of surface and sub-surface components, gravity effects, bit type, drill-pipe and BHA stiffness, damping coefficients, torsional stiffness, viscous damping, axial stiffness, Rayleigh/structural damping, lateral stiffness, and hydrodynamic damping coefficients. Additionally, boundary conditions related to the drill pipe and borehole wall contact, hydrodynamic and viscous damping forces due to mud drag, and the most important boundary is the bit-rock interactions that capture torque on the bit (*TOB*) and weight on the bit (*WOB*).

The lumped model offers a rough understanding of the dynamics occurring at various levels of the drill string, encompassing a range of degrees of freedom.

However, its accuracy declines at higher frequencies, especially as higher vibration modes become significant. This limitation results from the arbitrary choice of the number of degrees of freedom. Several lumped models have been developed to investigate the three possible modes in both coupled and uncoupled forms. An uncoupled model provides insights into the response of a single mode under specific conditions, while a lumped coupled model reveals key characteristics of the drill string, such as the coupled feature of vibration modes, energy transfer between vibrations modes, and the generative effects of each mode.

The discrete approach simplifies the understanding of drilling dynamics and offers low computational costs. This model is particularly effective in experimental test rigs (Wiercigroch et al., 2017; Kapitaniak et al., 2018; Wang et al., 2021), where identifiable lumped elements and a limited number of degrees of freedom are sufficient for a precise physical model of the tested mechanical system. Nonetheless, acknowledged disadvantages include varying simulation accuracy with the model's degrees of freedom, a significant change in the test results when a minor adjustment is made to the experimental parameters and overestimating distributed features, making it less applicable for analyzing full-size drilling systems.

In contrast, the distributed model offers a more accurate depiction of drill string behavior. Continuous models provide a more comprehensive analysis but require additional computational resources. A lumped parameter model can be considered a limiting case of a distributed model, where a quasi-static assumption is used for describing drill string dynamics. This assumption models drill pipes as a simple spring with constant torsional and axial stiffnesses, disregarding *BHA* flexibility and treating inertia as concentrated mass.

The lumped model cannot capture any resonance other than the fundamental frequency or replicate traveling waves within the structure. Therefore, it may not adequately represent the complex interactions in drill strings during actual operations.

For high-frequency operations, such as high-frequency torsional oscillations (*HFTOs*), higher-order distributed models are necessary to accurately capture *BHA* behavior. The dynamic behavior of drill string systems remains largely unpredictable due to its complexity and various uncertainties. Despite numerous studies, further research is needed to mitigate undesired vibrations. Key areas of uncertainty include bit/rock interactions, damping sources, pressure and temperature effects on components, and drill pipe straightness and eccentricity. Research like (Ying et al., 2021) using extended finite element methods (*XFEM*) for stress analysis and fracture mechanisms is scarce and critical for optimizing drill string design and operating conditions.

Accurately estimating axial, lateral, and torsional stiffnesses and damping ratios is vital, considering the complex role of drilling mud and other damping factors. Most experiments use simplified models instead of full-scale drill strings. Future studies should consider friction in sheaves, transitions between static and dynamic friction, and the effect of transitioning between different rock layers. *HFTOs*, for example, require advanced multiscale models to elucidate the mechanisms and correlations with various drilling dynamics, including whirling, bit bounce, stick-slip, and high-frequency axial oscillations (*HFAOs*). A comprehensive approach is crucial for advancing our understanding and improving our ability to manage these complex drilling dynamics.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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No data was used for the research described in the article.

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